

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

M2 Practical-8

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Class:TYIT	Batch:1
Date of Assignment: 30-9-2026	Date/Time of Submission: 30-9-2026

Aim: API Creation and Integration**A. Create a simple REST API using Express Steps:**

Create a folder & open that in vs code

Run -> npm init -y to create package.json file.

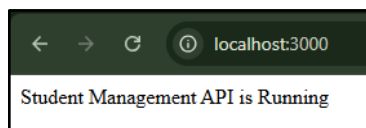
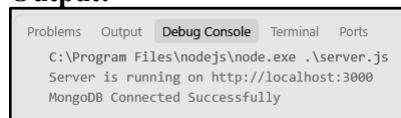
Add server.js file

Run command -> npm install mongoose express cors

Add following code into server.js file

Code:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB").then(() => {
  console.log("MongoDB Connected Successfully");
})
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);
});
```

Output:**B. Implement a GET API to retrieve data. Code:****1. Add following code practical (a) code before app.listen()**

```
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
```

Output:

C. Fetch data from the API in a React application.

Steps:

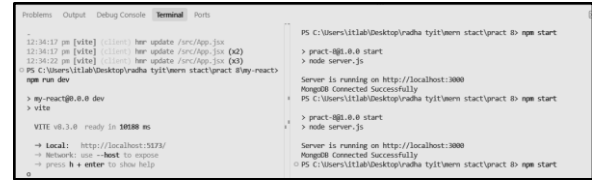
1. Create react project using following command
npm create vite@latest my-react
2. Choose react, javascript, ESLint
3. In terminal move to my-react & install axios npm
install axios
4. Add following code to App.jsx

Code:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  useEffect(() => {
    axios.get("http://localhost:3000/students") .then((response)
=> {setStudents(response.data);})
    .catch((error) => {console.log(error);}); }, []);
  return (
    <div>
      <h1>Student List</h1>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
```

export default App;

- 5) Split terminal & start server in that terminal. To start execute server.js using command node server.js.
- 6) Then in 1st terminal execute react application using npm run dev



Output:

Roll No	Name	Course	Marks
102	abhi	IT	96
101	radha	IT	96
103	radhika	IT	92

D. Send data from React to the server using a POST request.

E. Store data in MongoDB.

F. Display stored data in the React page.

Steps:

- 1) Replace App.jsx code.

Code:

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios
      .get("http://localhost:3000/Stud")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };
  useEffect(() => {
    getStudents();
  }, []);
  // Add student
  const addStudent = () => {
    const newStudent = {
      rollno: Number(rollNo),
      name: name,
```

```

    course: course,
    marks: Number(marks),
  };
  axios
    .post("http://localhost:3000/students", newStudent)
    .then((response) => {
      alert(response.data.message);
      setRollNo("");
      setName("");
      setCourse("");
      setMarks("");
      getStudents();
    })
    .catch((error) => {
      console.log(error);
    });
  };
  return (
    <div>
      <h1>Student Management System</h1>
      <h2>Add Student</h2>
      <input
        type="number"
        placeholder="Roll No"
        value={rollNo}
        onChange={(e) => setRollNo(e.target.value)}
      />
      <br />
      <br />
      <input
        type="text"
        placeholder="Student Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <br />
      <br />
      <input
        type="text"
        placeholder="Course"
        value={course}
        onChange={(e) => setCourse(e.target.value)}
      />
      <br />
      <br />
      <input
        type="number"
        placeholder="Marks"
        value={marks}
        onChange={(e) => setMarks(e.target.value)}
      />
      <br />
      <br />
      <button onClick={addStudent}>

```

```

    Add Student
  </button>
  <hr />
  <h2>Student List</h2>
  <table border="1">
    <thead>
      <tr>
        <th>Roll No</th>
        <th>Name</th>
        <th>Course</th>
        <th>Marks</th>
      </tr>
    </thead>
    <tbody>
      {students.map((student) => (
        <tr key={student._id}>
          <td>{student.rollno}</td>
          <td>{student.name}</td>
          <td>{student.course}</td>
          <td>{student.marks}</td>
        </tr>
      ))}
    </tbody>
  </table>
</div>
);
}
export default App;

```

2) Add the post method in the server.js file above app.listen().

Code:

```

const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Home Route
app.get("/", (req, res) => {

```

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```
res.send("Student Management API is Running");
});
// GET - Fetch all students
app.get("/Stud", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("Stud")
      .find()
      .toArray();
    res.json(students);
  } catch (error) {
    console.log(error);
    res.status(500).json({
      message: "Error fetching students",
    });
  }
});
// POST - Add a new student
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    const result = await mongoose.connection.db
      .collection("Stud")
      .insertOne(student);

    res.json({
      message: "Student added successfully",
      student: {
        _id: result.insertedId,
        ...student,
      },
    });
  } catch (error) {
    console.log(error);

    res.status(500).json({
      message: "Error adding student",
    });
  }
});
// Start Server
app.listen(PORT, () => {
  console.log(`Server is running on
http://localhost:${PORT}`);
});
```

3) Now 1st execute server.js then app.jsx in different Terminals.

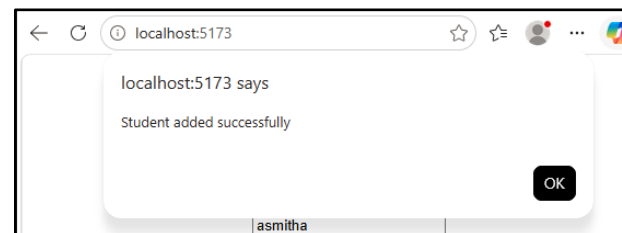
Output:

Student Management System

Add Student

Add Student

Roll No	Name	Course	Marks
102	abhi	IT	96
101	radha	IT	96
103	radhika	IT	92



Roll No	Name	Course	Marks
102	abhi	IT	96
101	radha	IT	96
103	radhika	IT	92
3	asmitha	CS	98